

ACHINTYA RAI

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EDUCATION

University of California, San Diego

Bachelor of Science in Computer Engineering

Sept. 2024 – June 2028

GPA: 3.9/4.0

- Competitive Programming Club, NSF REU Fellow (National Science Foundation), UC Scholars Research Fellow

WORK EXPERIENCE

Adobe

Software Engineering Intern

San Jose, CA

June 2026 – Sept. 2026

- Designed a time-series anomaly detector for payment volume, iterating over statistical algorithms to select a machine learning forecasting model handling multi-layer seasonality across **10+** workflows and **30+** countries.
- Developed SQL pipelines in Databricks to join two production event tables; designed an MLOps pipeline to schedule weekly retraining and hourly inference across all machine learning forecasting models.

VALT Health

Founding Software Engineer

Remote

Aug. 2025 – Feb. 2026

- Engineered a scalable Flask backend to handle **100k+** datapoints from fitness devices (Fitbit, Apple Watch); designed database schemas to unify data formats across **20+** distinct sources.
- Implemented concurrency-based batch processing and storage workflows in GCP, yielding a **35%** increase in execution speed and **60%** reduction in redundant database write operations.

Decklar (formerly Roambee)

Software Engineering Intern

Sunnyvale, CA

June 2024 – Aug. 2024

- Developed real-time shipment monitoring for **Amazon, Coca-Cola, and Samsung**, optimizing Python and NumPy pipelines to sustain large-scale systems processing **1M+** weekly events with a **15%** throughput boost.
- Implemented advanced data structures and algorithms for visualization and analytics modules, yielding **20%** faster reporting speeds and improved data integrity across internal shipment dashboards.

RESEARCH

Graph Theory & Neural Architecture

Undergraduate Research Fellow (NSF REU)

San Diego, CA

March 2026 – Present

- Engineered a Python pipeline to analyze artificial intelligence model embedding manifolds (Llama-2, DeepSeek-V2), implementing weight-matrix extraction utilities to reduce data footprints from **500GB** checkpoints to **13GB** localized manifolds.
- Optimized architectural placement algorithms and normalized preprocessing, achieving a **30x runtime reduction** (30 min → <1 min) while enabling large-scale manifold analysis on resource-constrained **8GB RAM** systems.

PROJECTS

NanoGPT | Python, PyTorch, C++, Transformers, LLM Optimization | [🔗](#)

Mar. 2026

- Implemented a GPT-style transformer from scratch in **Python** and **PyTorch**, hand-coding machine learning algorithms including multi-headed attention, causal masking, and backpropagation layers.
- Optimized model memory layout and compiled high-performance sub-modules in **C++**, reducing per-token latency by **25%** on resource-constrained systems.

Watchdog | Python, Gemini API, Browser-Use, Asyncio, SQLite [🔗](#)

April 2026

- Engineered AI-integrated software using the **browser-use SDK** and **Gemini Vision** to autonomously monitor browser tabs and intervene on distractions, delivering a fully functional agent system at DiamondHack 2026.
- Developed a concurrent CLI with **Asyncio** to manage real-time URL blacklisting and **Notion API** task synchronization, eliminating context-switching overhead across active browser sessions.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL (Postgres/SQLite), Bash, HTML/CSS

Development: React, Django, Flask, FastAPI, Asyncio, WebSockets, Node.js, JUnit, PyTest

AI & Cloud: PyTorch, machine learning, AWS, GCP, Gemini/OpenAI API, Docker, GitHub Actions, Linux